

Phase Number	Phase Name	Headlines	Technical Steps	Responsible	Duration	Status	Comments	
1	On-Premises Discovery Before Migration	Discover, document and validate the existing on-premises environment prior to migration	Run AWS Application Discovery Service on all source servers & source DBs					
			Document web server inventory: OS, CPU, RAM and disk layout Document DB server: engine, version and total schema size Document firewall rule sets and WAF policies Validate VPN connectivity to AWS Validate DNS agent prerequisites on web servers Validate MGN agent prerequisites on web servers Validate DNS prerequisites			2 days		
2	AWS Infrastructure Setup (Hub & Spoke)	Hub VPC & Core Networking Setup	Create Hub VPC with defined CIDR block Create Hub subnets: Public, NFW and Private across 2 AZ Create and attach Internet Gateway to Hub VPC Create NAT Gateway in Hub public subnet Configure Route Tables for Hub subnets Create Security Groups for ALB, NFW and Bastion Configure Network ACLs for subnet-level security			2 days		
		Application Load Balancer (ALB) & AWS WAF Setup	Create internet-facing ALB in Hub public subnets across 2 AZs Configure HTTP Listener with redirect to HTTPS Request SSL/TLS certificate in AWS Certificate Manager Create ALB Target Group pointing to NLB static IPs Create AWS WAF Web ACL with managed rule sets Associate WAF Web ACL with ALB			2 days		
		Spoke VPC Setup	Create Spoke VPC with defined CIDR block Create subnets: Private/NLB and Private/Web across 2 AZs Create DB Spoke subnets: Primary and Standby across 2 AZs Configure Security Groups			1 days		
		Transit Gateway (TGW) & Inter-VPC Routing	Configure Route Tables Create Transit Gateway in primary region Attach Hub VPC to Transit Gateway Attach Spoke VPC to Transit Gateway Configure VPC Route Tables and route propagations Enable Appliance mode in Hub attachment			1 days		
		AWS Network Firewall (NFW) Deployment	Create AWS Network Firewall in dedicated subnet within Hub VPC Create Gateways: Hub Group for web traffic Create Stateful Rule Group with IDS/IPS rules Create NFW Firewall Policy and attach rule groups Update Route Tables to route traffic through NFW			2 days		
		Bastion Host Setup in Hub	Launch Bastion EC2 instance in Hub public management subnet Configure Bastion Security Group to allow SSH from admin IPs only Test Bastion connectivity to Web and DB Spoke instances			1 days		
		On-Premises to AWS Site-to-Site VPN Connection	Create Customer Gateway (CGW) using on-premises firewall / router public IP Create Site-to-Site VPN Connection attached directly to Transit Gateway Download VPN configuration and apply to on-premises firewall/router Verify BGP sessions over the VPN tunnels to TGW Associate VPN attachment with the appropriate TGW route table Propagate on-premises routes to Hub/Spoke VPC route tables via TGW			2 days		
		AWS MGN Agent Installation & Replication (Web Servers)	Test end-to-end connectivity from on-premises to AWS resources across all spokes Enable AWS MGN in primary region Install AWS Replication Agent on source Linux web server Configure MGN staging area and replication settings Monitor initial replication sync in MGN console Configure MGN Launch Template for target EC2 instance Map source disks to target EBS volumes Configure post-launch actions for target instances Launch MGN test instance and verify DB boot Validate web application on MGN test instance Initiate target EC2 instance using Run application smoke tests on MGN test instance Install CloudWatch Agent on migrated web servers Verify SSL/TLS termination on ALB and InnoDB Update application configuration for AWS environment Create golden AMI from validated web instance Provision target DB in DB Spoke private subnet Configure DB Security Group Set up AWS DMS replication instance Create DMS source and target endpoints Create DMS replication task: Full Load + CDC Enable CDC for continuous replication Test application read/write operations against target DB Configure automated DB backups Configure DB parameter groups to match source settings Configure RDS standby instance in multi-AZ Enable DB encryption at rest and in transit Test DB failover procedure Obtain customer sign-off and prepare DB cutover plan Create EC2 Launch Template from validated web server AMI Create Auto Scaling Group for Web Servers across 2 AZs Configure ASG scaling policy Set ASG minimum, desired and maximum instance counts Create internal NLB in private subnet across 2 AZs Configure NLB listeners for HTTPS traffic Create NLB Target Group with health checks Register ASG with NLB Target Group Enable cross-zone load balancing on NLB Update ALB Target Group in Hub to point to NLB Run final end-to-end smoke tests and migrate endpoints to AWS environment Confirm DMS replication lag is at zero and validate complete data sync on target DB Verify all CloudWatch health checks, Route 53 health checks and alarms are passing Obtain customer sign-off to proceed with MGN cutover launch and DMS task termination			5 days		
		Web Server Cut-Over Preparation & Validation	Initiate target EC2 instance using Run application smoke tests on MGN test instance Install CloudWatch Agent on migrated web servers Verify SSL/TLS termination on ALB and InnoDB Update application configuration for AWS environment Create golden AMI from validated web instance Provision target DB in DB Spoke private subnet Configure DB Security Group Set up AWS DMS replication instance Create DMS source and target endpoints Create DMS replication task: Full Load + CDC Enable CDC for continuous replication Test application read/write operations against target DB Configure automated DB backups Configure DB parameter groups to match source settings Configure RDS standby instance in multi-AZ Enable DB encryption at rest and in transit Test DB failover procedure Obtain customer sign-off and prepare DB cutover plan Create EC2 Launch Template from validated web server AMI Create Auto Scaling Group for Web Servers across 2 AZs Configure ASG scaling policy Set ASG minimum, desired and maximum instance counts Create internal NLB in private subnet across 2 AZs Configure NLB listeners for HTTPS traffic Create NLB Target Group with health checks Register ASG with NLB Target Group Enable cross-zone load balancing on NLB Update ALB Target Group in Hub to point to NLB Run final end-to-end smoke tests and migrate endpoints to AWS environment Confirm DMS replication lag is at zero and validate complete data sync on target DB Verify all CloudWatch health checks, Route 53 health checks and alarms are passing Obtain customer sign-off to proceed with MGN cutover launch and DMS task termination			3 days		
		Database Migration Setup	Provision target DB in DB Spoke private subnet Configure DB Security Group Set up AWS DMS replication instance Create DMS source and target endpoints Create DMS replication task: Full Load + CDC Enable CDC for continuous replication Test application read/write operations against target DB Configure automated DB backups Configure DB parameter groups to match source settings Configure RDS standby instance in multi-AZ Enable DB encryption at rest and in transit Test DB failover procedure Obtain customer sign-off and prepare DB cutover plan Create EC2 Launch Template from validated web server AMI Create Auto Scaling Group for Web Servers across 2 AZs Configure ASG scaling policy Set ASG minimum, desired and maximum instance counts Create internal NLB in private subnet across 2 AZs Configure NLB listeners for HTTPS traffic Create NLB Target Group with health checks Register ASG with NLB Target Group Enable cross-zone load balancing on NLB Update ALB Target Group in Hub to point to NLB Run final end-to-end smoke tests and migrate endpoints to AWS environment Confirm DMS replication lag is at zero and validate complete data sync on target DB Verify all CloudWatch health checks, Route 53 health checks and alarms are passing Obtain customer sign-off to proceed with MGN cutover launch and DMS task termination			2 days		
Database Migration Execution & Validation	Test application read/write operations against target DB Configure automated DB backups Configure DB parameter groups to match source settings Configure RDS standby instance in multi-AZ Enable DB encryption at rest and in transit Test DB failover procedure Obtain customer sign-off and prepare DB cutover plan Create EC2 Launch Template from validated web server AMI Create Auto Scaling Group for Web Servers across 2 AZs Configure ASG scaling policy Set ASG minimum, desired and maximum instance counts Create internal NLB in private subnet across 2 AZs Configure NLB listeners for HTTPS traffic Create NLB Target Group with health checks Register ASG with NLB Target Group Enable cross-zone load balancing on NLB Update ALB Target Group in Hub to point to NLB Run final end-to-end smoke tests and migrate endpoints to AWS environment Confirm DMS replication lag is at zero and validate complete data sync on target DB Verify all CloudWatch health checks, Route 53 health checks and alarms are passing Obtain customer sign-off to proceed with MGN cutover launch and DMS task termination			3 days				
3	Migration & DR Implementation	Auto Scaling Group (ASG) for Web Servers in Spoke	Test DB failover procedure Obtain customer sign-off and prepare DB cutover plan Create EC2 Launch Template from validated web server AMI Create Auto Scaling Group for Web Servers across 2 AZs Configure ASG scaling policy Set ASG minimum, desired and maximum instance counts Create internal NLB in private subnet across 2 AZs Configure NLB listeners for HTTPS traffic Create NLB Target Group with health checks Register ASG with NLB Target Group Enable cross-zone load balancing on NLB Update ALB Target Group in Hub to point to NLB Run final end-to-end smoke tests and migrate endpoints to AWS environment Confirm DMS replication lag is at zero and validate complete data sync on target DB Verify all CloudWatch health checks, Route 53 health checks and alarms are passing Obtain customer sign-off to proceed with MGN cutover launch and DMS task termination			1 days		
		Network Load Balancer (NLB) Setup in Spoke	Configure ASG scaling policy Set ASG minimum, desired and maximum instance counts Create internal NLB in private subnet across 2 AZs Configure NLB listeners for HTTPS traffic Create NLB Target Group with health checks Register ASG with NLB Target Group Enable cross-zone load balancing on NLB Update ALB Target Group in Hub to point to NLB Run final end-to-end smoke tests and migrate endpoints to AWS environment Confirm DMS replication lag is at zero and validate complete data sync on target DB Verify all CloudWatch health checks, Route 53 health checks and alarms are passing Obtain customer sign-off to proceed with MGN cutover launch and DMS task termination			1 days		
		Production Cutover Validation	Enable cross-zone load balancing on NLB Update ALB Target Group in Hub to point to NLB Run final end-to-end smoke tests and migrate endpoints to AWS environment Confirm DMS replication lag is at zero and validate complete data sync on target DB Verify all CloudWatch health checks, Route 53 health checks and alarms are passing Obtain customer sign-off to proceed with MGN cutover launch and DMS task termination			2 days		
		DNS & Route 53 Configuration	Create Route 53 hosted zone for customer domain Link customer domain with Route 53 Configure Route 53 health check on primary ALB Create Route 53 failover records for primary ALB Set appropriate TTL values on all records Create private hosted zone for internal AWS endpoints Enable AWS Elastic Disaster Recovery in DR region Install DNS Replication Agent on source web servers Configure DNS staging area and replication settings Monitor DNS replication sync for web servers Create RDS Cross-Region Read Replica in DR region Monitor RDS cross-region replication lag Configure DNS Launch Settings for web servers in DR region Test promotion of RDS Cross-Region Read Replica to standalone DB Confirm DNS RPC meets agreed target (web servers) and RDS replica lag meets agreed target			1 days		
		AWS Elastic Disaster Recovery (DRG) & RDS Cross-Region Replica	Create DR Hub VPC in secondary region Create DR Spoke VPC in secondary region Create Transit Gateway in DR region and attach all public VPCs Deploy AWS Network Firewall in DR Hub VPC Deploy ALB and WAF in DR Hub VPC Create NLB in DR Spoke VPC Update Route 53 secondary failover routing record to point to DR site ALB endpoint Test DNS failover to DR endpoint Test DR failover and fallback runbook Conduct DR drill and obtain customer sign-off Enable AWS Security Hub in all regions Enable Amazon GuardDuty in primary and DR regions Enable AWS Config with compliance rules Enable AWS Macie for S3 bucket Enable AWS CloudTrail in all regions Configure S3 bucket policies and block public access Review and harden all Security Group rules Finalize and tune WAF rules Create AWS Backup Vault in primary region Create AWS Backup Vault in DR region Create Backup Plan for web server EC2 instances Create Backup Plan for DB instances Assign web server resources to Backup Plan Assign DB resources to Backup Plan Configure cross-region backup copy to DR region Test restore of EC2 web server from backup Test restore of DB from backup Configure Backup notification via SNS Document backup retention policies Verify backup compliance reports Configure Backup job monitoring in CloudWatch Generate Backup audit report Test complete restore scenario for web and DB Obtain customer sign-off on backup and restore procedures			5 days		
		DR Region Infrastructure Setup (Mirror of Primary)	Create DR Hub VPC in secondary region Create DR Spoke VPC in secondary region Create Transit Gateway in DR region and attach all public VPCs Deploy AWS Network Firewall in DR Hub VPC Deploy ALB and WAF in DR Hub VPC Create NLB in DR Spoke VPC Update Route 53 secondary failover routing record to point to DR site ALB endpoint Test DNS failover to DR endpoint Test DR failover and fallback runbook Conduct DR drill and obtain customer sign-off Enable AWS Security Hub in all regions Enable Amazon GuardDuty in primary and DR regions Enable AWS Config with compliance rules Enable AWS Macie for S3 bucket Enable AWS CloudTrail in all regions Configure S3 bucket policies and block public access Review and harden all Security Group rules Finalize and tune WAF rules Create AWS Backup Vault in primary region Create AWS Backup Vault in DR region Create Backup Plan for web server EC2 instances Create Backup Plan for DB instances Assign web server resources to Backup Plan Assign DB resources to Backup Plan Configure cross-region backup copy to DR region Test restore of EC2 web server from backup Test restore of DB from backup Configure Backup notification via SNS Document backup retention policies Verify backup compliance reports Configure Backup job monitoring in CloudWatch Generate Backup audit report Test complete restore scenario for web and DB Obtain customer sign-off on backup and restore procedures			5 days		
		Security Hardening & Compliance	Enable AWS Security Hub in all regions Enable Amazon GuardDuty in primary and DR regions Enable AWS Config with compliance rules Enable AWS Macie for S3 bucket Enable AWS CloudTrail in all regions Configure S3 bucket policies and block public access Review and harden all Security Group rules Finalize and tune WAF rules Create AWS Backup Vault in primary region Create AWS Backup Vault in DR region Create Backup Plan for web server EC2 instances Create Backup Plan for DB instances Assign web server resources to Backup Plan Assign DB resources to Backup Plan Configure cross-region backup copy to DR region Test restore of EC2 web server from backup Test restore of DB from backup Configure Backup notification via SNS Document backup retention policies Verify backup compliance reports Configure Backup job monitoring in CloudWatch Generate Backup audit report Test complete restore scenario for web and DB Obtain customer sign-off on backup and restore procedures			5 days		
		AWS Backup Configuration	Test restore of EC2 web server from backup Test restore of DB from backup Configure Backup notification via SNS Document backup retention policies Verify backup compliance reports Configure Backup job monitoring in CloudWatch Generate Backup audit report Test complete restore scenario for web and DB Obtain customer sign-off on backup and restore procedures			3 days		
		End-to-End Monitoring, Alerting & Observability Setup	Create CloudWatch dashboards for EC2, ALB, NLB and DB Configure CloudWatch Alarms for key metrics Set up CloudWatch log groups for application and system logs Configure SNS topics for alert notifications Test all alarms and verify notification delivery			2 days		
		5	Monitoring, Alerting & Observability	End-to-End Monitoring, Alerting & Observability Setup	Create CloudWatch dashboards for EC2, ALB, NLB and DB Configure CloudWatch Alarms for key metrics Set up CloudWatch log groups for application and system logs Configure SNS topics for alert notifications Test all alarms and verify notification delivery		2 days	